

Gauss Elimination Linear System

A linear system of 'n' equations in 'n' unknown $x_1, x_2, x_3, \dots, x_n$ is a set of equations

$E_1, E_2, \dots, E_n$ of the form.

$$\left. \begin{array}{l} E_1 : a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n = b_1 \\ E_2 : a_{21}x_1 + a_{22}x_2 + \dots + a_{2n}x_n = b_2 \\ \vdots \\ E_n : a_{n1}x_1 + a_{n2}x_2 + \dots + a_{nn}x_n = b_n \end{array} \right\} \rightarrow \textcircled{1}$$

where coefficients a_{jk} and b_j are given numbers.

If all the b_j 's are zero's the system is called homogeneous. otherwise it is called non-homogeneous.

We can write the above system as a single vector

$$AX = b \rightarrow \textcircled{2}$$

the coefficient matrix $A = [a_{jk}]$ is the $n \times n$ matrix.

$$A = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \dots & a_{nn} \end{bmatrix} \quad X = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} \quad b = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{bmatrix}$$

Gauss Elimination

This is a standard method for solving linear system (1). A systematic process of elimination that reduces (1) to 'triangular form'. Then system then can be easily solved using "back

Example

$$8x_2 + 2x_3 = -7$$

$$3x_1 + 5x_2 + 2x_3 = 8$$

$$6x_1 + 2x_2 + 8x_3 = 26$$

Solution

Interchange R_3 and R_1

$$6x_1 + 2x_2 + 8x_3 = 26$$

$$3x_1 + 5x_2 + 2x_3 = 8$$

$$8x_2 + 2x_3 = -7$$

$$\left[\begin{array}{ccc|c} 6 & 2 & 8 & 26 \\ 3 & 5 & 2 & 8 \\ 8 & 2 & 2 & -7 \end{array} \right]$$

$$R_2 - R_1/2$$

$$\left[\begin{array}{ccc|c} 6 & 2 & 8 & 26 \\ 0 & 4 & -2 & -5 \\ 0 & 8 & 2 & -7 \end{array} \right]$$

Interchange R_2 and R_3

$$\left[\begin{array}{ccc|c} 6 & 2 & 8 & 26 \\ 0 & 8 & 2 & -7 \\ 0 & 4 & -2 & -5 \end{array} \right]$$

$$R_3 - R_2/2$$

$$\left[\begin{array}{ccc|c} 6 & 2 & 8 & 26 \\ 0 & 8 & 2 & -7 \\ 0 & 0 & -3 & -3/2 \end{array} \right]$$

$$6x_1 + 2x_2 + 8x_3 = 26$$

$$8x_2 + 2x_3 = -7$$

$$-3x_3 = -3/2 \Rightarrow x_3 = 1/2$$

Using backward substitution

$$x_2 = \frac{1}{8}(-7 - 2x_3) = -1$$

$$x_1 = \frac{1}{6}(26 - 2x_2 - 8x_3) = 4$$

LU Factorization

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We discuss three related methods that are modifications of the Gauss Elimination, which require fewer arithmetic operations.

They are named after

Doolittle

~~Croft~~ Crout and

Cholesky.

and use the idea of the LU factorization of A.

$$A = LU \rightsquigarrow \textcircled{3}$$

where L is a lower triangular and U is the upper triangular matrix.

Example

$$A = \begin{bmatrix} 2 & 3 \\ 8 & 5 \end{bmatrix} = LU = \begin{bmatrix} 1 & 0 \\ 4 & 1 \end{bmatrix} \begin{bmatrix} 2 & 3 \\ 0 & -7 \end{bmatrix}$$

We can use LU Factorization for solving $AX = b$ in two steps, i.e.

$$AX = b$$

$$LUX = b$$

$$Ly = b \rightsquigarrow \textcircled{a}$$

$$UX = y \rightsquigarrow \textcircled{b}.$$

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Example

$$3x_1 + 5x_2 + 2x_3 = 8$$

$$0x_1 + 8x_2 + 2x_3 = -7$$

$$6x_1 + 2x_2 + 8x_3 = 26$$

Solution

$$\begin{bmatrix} 3 & 5 & 2 \\ 0 & 8 & 2 \\ 6 & 2 & 8 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ m_{21} & 1 & 0 \\ m_{31} & m_{32} & 1 \end{bmatrix} \begin{bmatrix} u_{11} & u_{12} & u_{13} \\ 0 & u_{22} & u_{23} \\ 0 & 0 & u_{33} \end{bmatrix}$$

$$a_{11} = 3 = u_{11}$$

$$a_{12} = 5 = u_{12}$$

$$a_{13} = 2 = u_{13}$$

$$a_{21} = 0 = m_{21} \times u_{11}$$

$$a_{22} = 8 = m_{21}u_{12} + u_{22}$$

$$a_{23} = 2 = m_{21}u_{13} + u_{23}$$

$$m_{21} = 0$$

$$u_{22} = 8$$

$$u_{23} = 2$$

$$a_{31} = 6 = m_{31}u_{11}$$

$$a_{32} = 2 = m_{31}u_{12} + m_{32}u_{22}$$

$$a_{33} = 8 = m_{31}u_{13} + m_{32}u_{23} + u_{33}$$

$$m_{31} = 2$$

$$m_{32} = -1$$

$$u_{33} = 6$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 2 & -1 & 1 \end{bmatrix} \begin{bmatrix} 3 & 5 & 2 \\ 0 & 8 & 2 \\ 0 & 0 & 6 \end{bmatrix}$$

L

U

$$Ly = b$$

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$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 2 & -1 & 1 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = \begin{bmatrix} 8 \\ -7 \\ 26 \end{bmatrix}$$

$$y_1 = 8$$

$$y_2 = -7$$

$$y_3 = 3$$

~~Ux = y~~

$$Ux = y$$

$$\begin{bmatrix} 3 & 5 & 2 \\ 0 & 8 & 2 \\ 0 & 0 & 6 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 8 \\ -7 \\ 3 \end{bmatrix}$$

$$x_3 = 1/2$$

$$x_2 = -1$$

$$x_1 = 4$$

Our example suggest for general n entries of the matrices L and U in the Doolittle method are computed from

$$u_{1k} = a_{1k} \quad k = 1, 2, \dots, n$$

$$m_{j1} = \frac{a_{j1}}{u_{11}} \quad j = 2, 3, \dots, n$$

$$u_{jk} = a_{jk} - \sum_{s=1}^{j-1} m_{js} u_{sk} \quad k = j, \dots, n \quad j \geq 2$$

$$m_{jk} = \frac{1}{u_{kk}} \left(a_{jk} - \sum_{s=1}^{k-1} m_{js} u_{sk} \right) \quad j = k+1, \dots, n \quad k \geq 2.$$

Doolittle's Method

If 'L' (Lower diagonal Matrix) in LU-Factorization have main diagonal elements 1's then this is called Doolittle's Method.

Crout's Method

If 'U' (Upper diagonal Matrix) in

LU-Factorization have main diagonal elements 1's then this is called Crout's Method.

Cholesky's Method

For symmetric ($A = A^T$) and

Positive definite matrix ($\underline{x^T A x} \geq 0$) $\forall x \neq 0$ we can

choose $U = L^T$ in LU-Factorization i.e.

$$\begin{aligned} A &= LU \\ &= L L^T \end{aligned}$$

Thus $u_{jk} = m_{kj}$ (but cannot impose conditions on the main

diagonal elements)

example

$$A = \begin{bmatrix} 4 & 2 & 14 \\ 2 & 17 & -5 \\ 14 & -5 & 83 \end{bmatrix} = L L^T = \begin{bmatrix} m_{11} & 0 & 0 \\ m_{21} & m_{22} & 0 \\ m_{31} & m_{32} & m_{33} \end{bmatrix} \begin{bmatrix} m_{11} & m_{21} & m_{31} \\ 0 & m_{22} & m_{32} \\ 0 & 0 & m_{33} \end{bmatrix}$$

$$a_{11} = m_{11} \cdot m_{11} = 4$$

$$m_{11}^2 = 4$$

$$m_{11} = 2$$

$$a_{12} = m_{11} m_{21} = 2$$

$$m_{21} = \frac{2}{2}$$

$$m_{21} = 1$$

$$a_{13} = m_{11} m_{31} = 14$$

$$m_{31} = \frac{a_{13}}{m_{11}} = \frac{14}{2} = 7$$

$$a_{22} = m_{21}^2 + m_{22}^2 = 17$$

$$m_{22}^2 = \frac{17 - m_{21}^2}{1} = \frac{17 - 1}{1}$$

$$m_{22}^2 = 16$$

$$m_{22} = 4$$

$$a_{31} = m_{31} m_{11} = 14$$

$$m_{31} = \frac{14}{m_{11}} = \frac{14}{2} = 7$$

$$a_{23} = m_{21} m_{31} + m_{22} m_{32} = -5$$

$$m_{32} = \frac{-5 - m_{21} m_{31}}{m_{22}} = \frac{-5 - (1 \times 7)}{4} = \frac{-5 - 7}{4} = \frac{-12}{4} = -3$$

$$a_{33} = m_{31}^2 + m_{32}^2 + m_{33}^2 = 83$$

$$m_{33}^2 = 83 - m_{31}^2 - m_{32}^2$$

$$m_{33} = \sqrt{83 - (7)^2 - (-3)^2}$$

$$= 5$$

$$A = \begin{bmatrix} 4 & 2 & 14 \\ 2 & 17 & -5 \\ 14 & -5 & 83 \end{bmatrix} = \begin{bmatrix} 2 & 0 & 0 \\ 1 & 4 & 0 \\ 7 & -3 & 5 \end{bmatrix} \begin{bmatrix} 2 & 1 & 7 \\ 0 & 4 & -3 \\ 0 & 0 & 5 \end{bmatrix}$$

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$$m_{11} = l_{11} = \sqrt{a_{11}}$$

$$m_{j1} = l_{j1} = \frac{a_{j1}}{l_{11}}$$

$$j = 2, \dots, n$$

$$m_{jj} = l_{jj} = \sqrt{a_{jj} - \sum_{s=1}^{j-1} l_{js}^2}$$

$$m_{pj} = l_{pj} = \frac{1}{l_{jj}} \left(a_{pj} - \sum_{s=1}^{j-1} l_{js} l_{ps} \right)$$

$$p = j+1, \dots, n; j \geq 2$$

Example Cholesky's Method

$$4x_1 + 2x_2 + 14x_3 = 14$$

$$2x_1 + 17x_2 - 5x_3 = -101$$

$$14x_1 - 5x_2 + 83x_3 = 155$$

Solution

$$\begin{bmatrix} 4 & 2 & 14 \\ 2 & 17 & -5 \\ 14 & -5 & 83 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 14 \\ -101 \\ 155 \end{bmatrix}$$

$A \quad X = b$

~~using~~ writing $A = L L^T$

$$\begin{bmatrix} 4 & 2 & 14 \\ 2 & 17 & -5 \\ 14 & -5 & 83 \end{bmatrix} = \begin{bmatrix} 2 & 0 & 0 \\ 1 & 4 & 0 \\ 7 & -3 & 5 \end{bmatrix} \begin{bmatrix} 2 & 1 & 7 \\ 0 & 4 & -3 \\ 0 & 0 & 5 \end{bmatrix}$$

Now solve $Ly = b$

$$\begin{bmatrix} 2 & 0 & 0 \\ 1 & 4 & 0 \\ 7 & -3 & 5 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = \begin{bmatrix} 14 \\ -101 \\ 155 \end{bmatrix}$$

$$y_1 = 7$$

$$y_2 = -27$$

$$y_3 = 5$$

Now $UX = Y \Rightarrow L^T X = Y$

$$\begin{bmatrix} 2 & 1 & 7 \\ 0 & 4 & -3 \\ 0 & 0 & 5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 7 \\ -27 \\ 5 \end{bmatrix}$$

$$X = \begin{bmatrix} 3 \\ -6 \\ 1 \end{bmatrix}$$

Ans.

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It can be used to accelerate the convergence for systems that are convergent by the Gauss-Seidel method.

Example

$$2x_1 + 8x_2 - x_3 = 11$$

$$5x_1 - x_2 + x_3 = 10$$

$$-x_1 + x_2 + 4x_3 = 3$$

Initial approximation

$$x^0 = [0, 0, 0]^T$$

$$\omega = 0.35$$

Solution

$$x_1^{k+1} = (1-\omega)x_1^k + \frac{\omega}{2} [11 - 8x_2^k + x_3^k]$$

$$x_2^{k+1} = (1-\omega)x_2^k + \frac{\omega}{-1} [10 - 5x_1^{k+1} + x_3^k]$$

$$x_3^{k+1} = (1-\omega)x_3^k + \frac{\omega}{4} [3 + x_1^{k+1} - x_2^{k+1}]$$

For $k=0$

$$x_1^1 = (1-0.35)x_1^0 + \frac{0.35}{2} [11 - 8x_2^0 + x_3^0] = 1.925000$$

$$x_2^1 = (1-0.35)x_2^0 + \frac{0.35}{-1} [10 - 5x_1^1 + x_3^0] = -0.131250$$

$$x_3^1 = (1-0.35)x_3^0 + \frac{0.35}{4} [3 + x_1^1 - x_2^1] = 0.442422$$

For $k=1$

$$x_1^2 = 3.437424$$

$$x_2^2 = 2.585027$$

$$x_3^2 = 0.624659$$

For $k=2$

$$x_1^3 = 0.649603$$

$$x_2^3 = -0.464296$$

$$x_3^3 = 0.765995$$

$$k=39$$

$$x_1^{40} = 2$$

$$x_2^{40} = 1$$

$$x_3^{40} = 1$$